

Total No. of Questions : 8]

[Total No. of Printed Pages : 2

Roll No

EE/EX-5003-CBGS

B.E. V Semester

Examination, December 2020

Choice Based Grading System (CBGS)

Switchgear and Protection

Time : Three Hours

Maximum Marks : 70

Note: i) Attempt any five questions.

ii) All questions carry equal marks.

1. a) Draw the connection of sequence networks for L-L-G fault and also derive the expression for fault current. 7
b) A 25 MVA, 13.2kV alternator with solidly grounded neutral has a subtransient reactance of 0.25 p.u. The negative and zero sequence reactances are 0.35 and 0.1 p.u. respectively. Determine the fault current and the line to line voltages at fault when a double line to ground fault occurs at the terminals of the alternator. 7
2. a) A 3-phase, 11kV, 25MVA generator with $X_0 = 0.05$ p.u.; $X_1 = 0.2$ p.u. and $X_2 = 0.2$ p.u. is grounded through a reactance of 0.3Ω . Calculate the fault current for a single line to ground fault. 7
b) Describe the principle of current limiting reactor. 7
3. a) Explain the construction, principle of operation and application of a directional over current relay. 7
b) What is universal torque equation? Using these equation derive the following characteristics. 7
 - i) Impedance relay
 - ii) Reactance relay
 - iii) Mho relay

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[2]

4. a) The current setting of a relay is 5 Amp. PSM = 1.5

$$\text{TMS} = 0.2, \text{CT ratio} = \frac{400}{5}; \text{fault current} = 6000 \text{ amp.}$$

Determine the operating time of relay. 7

At TMS=1, operative time at various PSM are:

PSM	2	4	5	8	10	20
Op. time Csec	10	5	4	3	2.8	2.4

- b) Explain the different types of over current relays depending upon the time of operation. 7

5. Discuss the construction, operating principle and application of SF₆ circuit Breaker. What are its advantages? For which voltage rating range it is used. 14

6. a) Describe Merz-price protection of transformer. 7

- b) A 3-phase transformer of 220/11,000 line volts is connected in star/delta. The protective transformers on 220V side have a current ratio of 600/5. What should be the C.T. ratio on 11,000 V side. 7

7. a) List the protection schemes employed for synchronous generator. Explain in short explaining the consequence of mal operation of these protections in each case. 7

- b) Explain protection scheme used for feeders. 7

8. Write short notes on (any two) : 2×7=14

- a) Protection against lightning
- b) Insulation co-ordination
- c) Surge absorber
- d) Earthing transformers
